

# ELITE IGCSE ACADEMY

*Student Past Paper Solutions*

## May 2021 P1H Worked Solutions

May 2021 | P1H

33 questions   ■   question image followed by worked solution

PREPARED BY

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**PAST PAPER QUESTION****Q#: 1****Marks: 3****TOPIC: Standard & Compound Units**

May 2021 P1H - May 2021 | P1H

- 1** A plane flew from Madrid to Dubai.

The distance the plane flew was 5658 km.

The flight time was 8 hours 12 minutes.

Work out the average speed of the plane.

..... km/h

**(Total for Question 1 is 3 marks)**

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**PAST PAPER SOLUTION****Q#: 1****Marks: 3****TOPIC: Standard & Compound Units**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Corrected to Standard & Compound Units.**Solution**

$$8 \text{ h } 12 \text{ min} = 8 + \frac{12}{60} = 8.2 \text{ hours}$$

$$\text{average speed} = \frac{5658}{8.2} = 690$$

**FINAL ANSWER**

690 km/h.

**PAST PAPER QUESTION****Q#: 2****Marks: 2**TOPIC: **Sequences**

May 2021 P1H - May 2021 | P1H

- 2 Here are the first 4 terms of an arithmetic sequence.

85      79      73      67

Find an expression, in terms of  $n$ , for the  $n$ th term of the sequence.

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(Total for Question 2 is 2 marks)

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**PAST PAPER SOLUTION****Q#: 2****Marks: 2**TOPIC: **Sequences**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Sequences. The tag is correct.**Solution**

The sequence is

$$85, 79, 73, 67$$

The common difference is  $-6$ .

$$\text{nth term} = 85 + (n - 1)(-6)$$

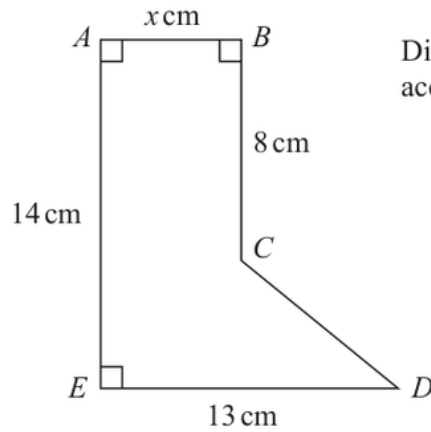
$$= 85 - 6n + 6 = 91 - 6n$$

**FINAL ANSWER**

$$91 - 6n.$$

**PAST PAPER QUESTION****Q#: 3****Marks: 4**TOPIC: **Area & Perimeter**

May 2021 P1H - May 2021 | P1H

**3**Diagram **NOT**  
accurately drawnThe diagram shows the shape  $ABCDE$ .The area of the shape is  $91.8\text{ cm}^2$ Work out the value of  $x$ . $x = \dots\dots\dots$ **(Total for Question 3 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 3****Marks: 4****TOPIC: Area & Perimeter**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Area & Perimeter. The tag is correct.**Solution**

Split the shape into a rectangle and a triangle.

Rectangle area:

$$14x$$

The triangle has base  $13 - x$  and height 6, so its area is

$$\frac{1}{2}(13 - x)(6) = 39 - 3x$$

Total area:

$$14x + 39 - 3x = 91.5$$

$$11x = 52.5$$

$$x = 4.772 \dots$$

**FINAL ANSWER**

4.77 cm.

**PAST PAPER QUESTION****Q#: 4****Marks: 3**TOPIC: **Ratio Toolkit**

May 2021 P1H - May 2021 | P1H

- 4 On a farm there are chickens, ducks and pigs.

The ratio of the number of chickens to the number of ducks is 7:2

The ratio of the number of ducks to the number of pigs is 5:9

There are 36 pigs on the farm.

Work out the number of chickens on the farm.

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(Total for Question 4 is 3 marks)

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**PAST PAPER SOLUTION****Q#: 4****Marks: 3**TOPIC: **Ratio Toolkit**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Ratio Toolkit. The tag is correct.**Method**

Ducks : pigs = 5 : 9.

There are 36 pigs, so

$$9 \text{ parts} = 36$$

$$1 \text{ part} = 4$$

$$\text{ducks} = 5 \times 4 = 20$$

Chickens : ducks = 7 : 2.

$$2 \text{ parts} = 20$$

$$1 \text{ part} = 10$$

$$\text{chickens} = 7 \times 10 = 70$$

**FINAL ANSWER**

70

**PAST PAPER QUESTION****Q#: 5****Marks: 6****TOPIC: Rearranging Formulas**

May 2021 P1H - May 2021 | P1H

- 5 (a) Expand and simplify  $3x(2x + 3) - x(3x + 5)$

.....  
(2)

- (b) Make  $t$  the subject of the formula  $p = at - d$

.....  
(2)

Given that  $\frac{w^5 \times w^n}{w^3} = w^{10}$

- (c) work out the value of  $n$ .

$n =$  .....  
(2)

---

**(Total for Question 5 is 6 marks)**

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**PAST PAPER SOLUTION****Q#: 5****Marks: 6****TOPIC: Rearranging Formulas**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Rearranging formulas. The tag is acceptable in this mixed algebra question.**Solution**

(a)

$$3x(2x + 3) - x(3x + 5) = 6x^2 + 9x - 3x^2 - 5x$$

$$= 3x^2 + 4x$$

(b)

$$p = at - d$$

$$p + d = at$$

$$t = \frac{p + d}{a}$$

(c)

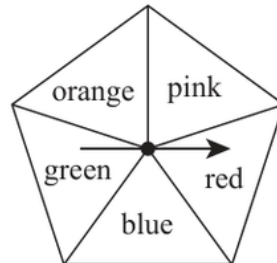
$$\frac{w^5 \times w^n}{w^3} = w^{n+2} = w^{10}$$

So  $n + 2 = 10$ , hence  $n = 8$ .**FINAL ANSWER**(a)  $3x^2 + 4x$ , (b)  $t = \frac{p+d}{a}$ , (c)  $n = 8$

**PAST PAPER QUESTION****Q#: 6****Marks: 5****TOPIC: Probability Toolkit**

May 2021 P1H - May 2021 | P1H

- 6 Grace has a biased 5-sided spinner.



Grace is going to spin the arrow on the spinner once.

The table below gives the probabilities that the spinner will land on red or on blue or on green.

| Colour      | Red  | Blue | Green | Orange | Pink |
|-------------|------|------|-------|--------|------|
| Probability | 0.20 | 0.12 | 0.08  |        |      |

The probability that the spinner will land on orange is 3 times the probability that the spinner will land on pink.

- (a) Work out the probability that the spinner will land on orange.

.....  
(3)

Grace spins the arrow on the spinner 150 times.

- (b) Work out an estimate for the number of times the spinner lands on blue.

.....  
(2)

**(Total for Question 6 is 5 marks)**

**PAST PAPER SOLUTION****Q#: 6****Marks: 5**TOPIC: **Probability Toolkit**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Probability Toolkit. The tag is correct.**Solution**

Let the probability of pink be  $x$ . Then the probability of orange is  $3x$ .

$$0.20 + 0.12 + 0.08 + x + 3x = 1$$

$$4x = 0.60$$

$$x = 0.15$$

So the probability of orange is

$$3x = 0.45$$

Expected blue sweets:

$$150 \times 0.12 = 18$$

**FINAL ANSWER**

pink 0.15, orange 0.45, blue 18.

**PAST PAPER QUESTION****Q#: 7****Marks: 4****TOPIC: Solving Inequalities**

May 2021 P1H - May 2021 | P1H

7  $-4 \leq 2y < 6$

$y$  is an integer.

(a) Write down all the possible values of  $y$ .

.....  
(2)

(b) Solve the inequality  $7t - 3 \leq 2t + 31$

Show your working clearly.

.....  
(2)

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**(Total for Question 7 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 7****Marks: 4****TOPIC: Solving Inequalities**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Solving inequalities. The tag is correct.**Solution**

(a)

$$-4 \leq 2y < 6$$

Divide by 2:

$$-2 \leq y < 3$$

Since  $y$  is an integer,

$$y = -2, -1, 0, 1, 2$$

(b)

$$7t - 3 \leq 2t + 31$$

$$5t \leq 34$$

$$t \leq \frac{34}{5}$$

**FINAL ANSWER** $y = -2, -1, 0, 1, 2$ , and  $t \leq \frac{34}{5}$ .

**PAST PAPER QUESTION****Q#: 8****Marks: 4****TOPIC: Powers, Roots & Standard Form**

May 2021 P1H - May 2021 | P1H

8 The table shows the populations of five countries.

| Country | Population        |
|---------|-------------------|
| China   | $1.4 \times 10^9$ |
| Germany | $8.2 \times 10^7$ |
| Sweden  | $9.9 \times 10^6$ |
| Fiji    | $9.1 \times 10^5$ |
| Malta   | $4.3 \times 10^5$ |

- (a) Work out the difference between the population of China and the population of Germany.  
Give your answer in standard form.

.....  
(2)

Given that

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

- (b) work out the value of  $k$ .  
Give your answer correct to the nearest whole number.

$k =$  .....  
(2)

**(Total for Question 8 is 4 marks)**



**PAST PAPER SOLUTION****Q#: 8****Marks: 4****TOPIC: Powers, Roots & Standard Form**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Powers, roots and standard form. The tag is correct.**Method**

For part (a),

$$1.4 \times 10^9 = 1400000000$$

$$8.2 \times 10^7 = 82000000$$

Difference:

$$1400000000 - 82000000 = 1318000000$$

$$1318000000 = 1.318 \times 10^9$$

For part (b),

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

So

$$k = \frac{\text{population of Sweden}}{\text{population of Fiji}}$$

$$k = \frac{9.9 \times 10^6}{9.1 \times 10^5} = 10.879 \dots$$

To the nearest whole number,

$$k = 11$$

**FINAL ANSWER**(a)  $1.318 \times 10^9$ , (b) 11

**PAST PAPER QUESTION****Q#: 9****Marks: 5**TOPIC: **Factorising**

May 2021 P1H - May 2021 | P1H

9 (a) Factorise fully  $25a^4c^7d + 45a^9c^3h$

.....  
(2)

(b) Solve  $(2x + 5)^2 = (2x + 3)(2x - 1)$

$x =$  .....  
(3)

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**(Total for Question 9 is 5 marks)**

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**PAST PAPER SOLUTION****Q#: 9****Marks: 5****TOPIC: Factorising**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Factorising. The tag is correct.**Solution**

(a)

$$25a^4c^7d + 45a^9c^3h = 5a^4c^3(5c^4d + 9a^5h)$$

(b)

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

$$4x^2 + 20x + 25 = 4x^2 + 4x - 3$$

$$16x = -28$$

$$x = -\frac{7}{4}$$

**FINAL ANSWER**(a)  $5a^4c^3(5c^4d + 9a^5h)$ , (b)  $x = -\frac{7}{4}$

**PAST PAPER QUESTION****Q#: 10****Marks: 3**TOPIC: **Statistics Toolkit**

May 2021 P1H - May 2021 | P1H

**10** Jethro has sat 5 tests.

Each test was marked out of 100 and Jethro's mean mark for the 5 tests is 74

Jethro has to sit one more test that is also to be marked out of 100

Jethro wants his mean mark for all 6 tests to be at least 77

Work out the least mark that Jethro needs to get for the last test.

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**(Total for Question 10 is 3 marks)**

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**PAST PAPER SOLUTION****Q#: 10****Marks: 3**TOPIC: **Statistics Toolkit**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

The total for the first 5 tests is

$$5 \times 74 = 370$$

For an average of at least 77 after 6 tests, the total must be at least

$$6 \times 77 = 462$$

So the last test score must be at least

$$462 - 370 = 92$$

**FINAL ANSWER**

at least 92.

**PAST PAPER QUESTION****Q#: 11****Marks: 4**TOPIC: **Algebraic Roots & Indices**

May 2021 P1H - May 2021 | P1H

**11**  $\sqrt{2} \times 16 = 2^x$

- (a) Find the value of  $x$ .  
Show your working clearly.

$$x = \dots\dots\dots$$

(2)

$$\frac{(11^{-6})^5}{11^4} = 11^n$$

- (b) Find the value of  $n$ .  
Show your working clearly.

$$n = \dots\dots\dots$$

(2)

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**(Total for Question 11 is 4 marks)**

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**PAST PAPER SOLUTION****Q#: 11****Marks: 4**TOPIC: **Algebraic Roots & Indices**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Algebraic Roots and Indices. The tag is correct.**Method**

$$\sqrt{2} \times 16 = 2^x$$

Write both factors as powers of 2:

$$\sqrt{2} = 2^{1/2}, \quad 16 = 2^4$$

So

$$2^{1/2} \times 2^4 = 2^{9/2}$$

Therefore

$$x = \frac{9}{2}$$

For part (b),

$$\frac{(11^{-6})^5}{11^4} = \frac{11^{-30}}{11^4} = 11^{-30-4} = 11^{-34}$$

So

$$n = -34$$

**FINAL ANSWER**(a)  $x = \frac{9}{2}$ , (b)  $n = -34$ .

**PAST PAPER QUESTION****Q#: 12****Marks: 2**TOPIC: **Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

- 12** The diagram shows a sector of a circle with radius 7 cm.

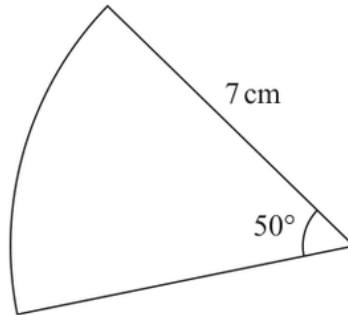


Diagram **NOT**  
accurately drawn

Work out the length of the arc of the sector.  
Give your answer correct to one decimal place.

..... cm

(Total for Question 12 is 2 marks)



**PAST PAPER SOLUTION****Q#: 12****Marks: 2****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Circles, Arcs & Sectors. The tag is correct.**Solution**

$$\begin{aligned}\text{arc length} &= \frac{50}{360} \times 2\pi(7) \\ &= 6.108 \dots\end{aligned}$$

**FINAL ANSWER**

6.1 cm.

**PAST PAPER QUESTION****Q#: 13****Marks: 3****TOPIC: Expanding Brackets**

May 2021 P1H - May 2021 | P1H

- 13** Expand and simplify  $4x(3x + 1)(2x - 3)$   
Show your working clearly.

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(Total for Question 13 is 3 marks)

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**PAST PAPER SOLUTION****Q#: 13****Marks: 3****TOPIC: Expanding Brackets**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Expanding brackets. The tag is correct.**Solution**

$$\begin{aligned}4x(3x + 1)(2x - 3) &= 4x(6x^2 - 7x - 3) \\ &= 24x^3 - 28x^2 - 12x\end{aligned}$$

**FINAL ANSWER**

$$24x^3 - 28x^2 - 12x$$

**PAST PAPER QUESTION****Q#: 14****Marks: 2**TOPIC: **Statistics Toolkit**

May 2021 P1H - May 2021 | P1H

**14** Here is the number of goals that Henri's team scored one summer in each water polo match.

5    8    9    11    13    13    14    15    16    17    20

Find the interquartile range of the numbers of goals.  
Show your working clearly.

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(Total for Question 14 is 2 marks)

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**PAST PAPER SOLUTION****Q#: 14****Marks: 2**TOPIC: **Statistics Toolkit**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

The goals in order are

5, 8, 9, 11, 13, 13, 14, 15, 16, 17, 20

$$Q_1 = 9, \quad Q_3 = 16$$

$$\text{IQR} = 16 - 9 = 7$$

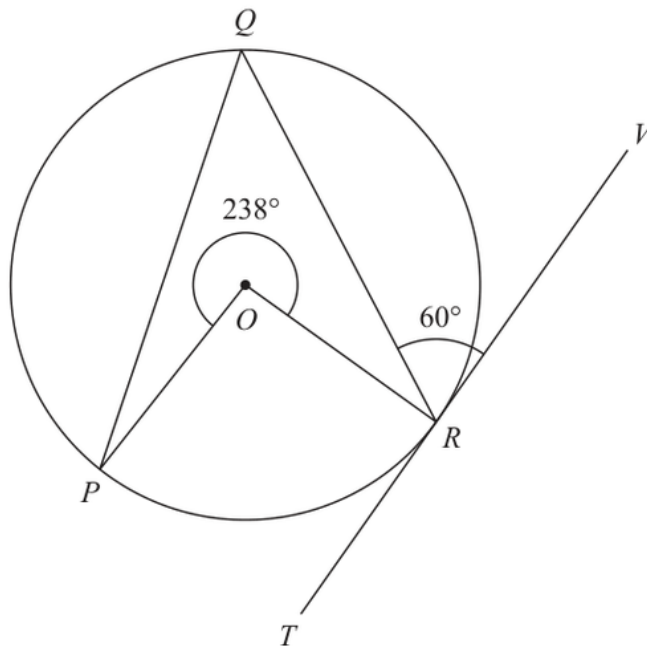
**FINAL ANSWER**

7.

**PAST PAPER QUESTION****Q#: 15****Marks: 4****TOPIC: Circle Theorems**

May 2021 P1H - May 2021 | P1H

- 15**  $P$ ,  $Q$  and  $R$  are points on a circle, centre  $O$ .  
 $TRV$  is the tangent to the circle at  $R$ .

Diagram **NOT**  
accurately drawnReflex angle  $POR = 238^\circ$ Angle  $QRV = 60^\circ$ Calculate the size of angle  $OPQ$ .

Give a reason for each stage of your working.

.....  
 (Total for Question 15 is 4 marks)

**PAST PAPER SOLUTION****Q#: 15****Marks: 4**TOPIC: **Circle Theorems**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Circle Theorems. The tag is correct.**Solution**The reflex angle  $POR$  is  $238^\circ$ , so the smaller angle is

$$360^\circ - 238^\circ = 122^\circ$$

Since  $OP = OR$ , triangle  $OPR$  is isosceles:

$$\angle OPR = \frac{180^\circ - 122^\circ}{2} = 29^\circ$$

By the alternate segment theorem,

$$\angle QPR = 60^\circ$$

Therefore

$$\angle QPO = 60^\circ - 29^\circ = 31^\circ$$

**FINAL ANSWER** **$31^\circ$ .**

**PAST PAPER QUESTION****Q#: 16****Marks: 2**TOPIC: **Algebraic Proof**

May 2021 P1H - May 2021 | P1H

**16** Use algebra to show that the recurring decimal  $0.28\dot{1}\dot{3} = \frac{557}{1980}$

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(Total for Question 16 is 2 marks)

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**PAST PAPER SOLUTION****Q#: 16****Marks: 2**TOPIC: **Algebraic Proof**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Algebraic proof. The tag is correct.**Solution**

Let

$$x = 0.28131313 \dots$$

Then

$$100x = 28.131313 \dots$$

and

$$10000x = 2813.131313 \dots$$

Subtract:

$$10000x - 100x = 2813.131313 \dots - 28.131313 \dots$$

$$9900x = 2785$$

$$x = \frac{2785}{9900} = \frac{557}{1980}$$

**FINAL ANSWER**

$$0.28131313 \dots = \frac{557}{1980}.$$

**PAST PAPER QUESTION****Q#: 17****Marks: 3**TOPIC: **Algebraic Proof**

May 2021 P1H - May 2021 | P1H

- 17** Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

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(Total for Question 17 is 3 marks)

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**PAST PAPER SOLUTION****Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Algebraic proof. The tag is correct.**Solution**

Let the three consecutive even numbers be

$$2n - 2, \quad 2n, \quad 2n + 2$$

The middle number is  $2n$ .

The difference between the square of the largest and the square of the smallest is

$$\begin{aligned} & (2n + 2)^2 - (2n - 2)^2 \\ &= (4n^2 + 8n + 4) - (4n^2 - 8n + 4) \\ &= 16n \end{aligned}$$

Since  $8 \times \text{middle number} = 8(2n) = 16n$ , the result is 8 times the middle number.**FINAL ANSWER**

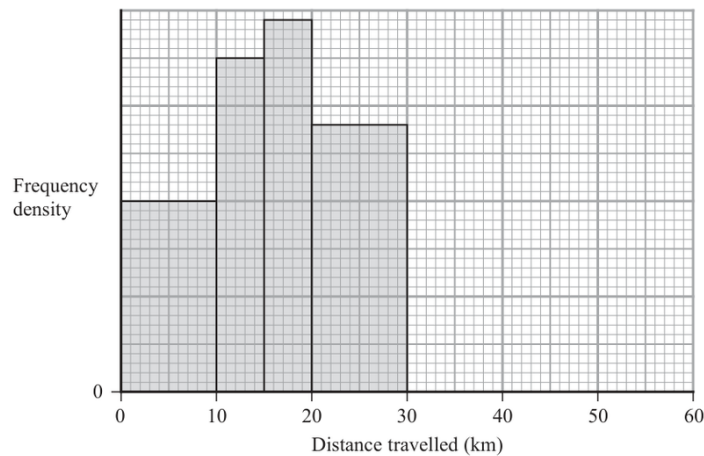
The difference is always 8 times the middle number.

**PAST PAPER QUESTION****Q#: 18****Marks: 5****TOPIC: Histograms**

May 2021 P1H - May 2021 | P1H

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

| Distance ( $d$ km) | Frequency |
|--------------------|-----------|
| $0 \leq d < 10$    | 40        |
| $10 \leq d < 15$   |           |
| $15 \leq d < 20$   |           |
| $20 \leq d < 30$   |           |
| $30 \leq d < 60$   | 30        |



Using the information in the table and in the histogram,

- (a) complete the table,

(2)

- (b) complete the histogram.

(1)

One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)

(Total for Question 18 is 5 marks)

**PAST PAPER SOLUTION****Q#: 18****Marks: 5**TOPIC: **Histograms**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Histograms. The tag is correct.**Solution**The  $0 \leq d < 10$  class has frequency 40, so its density is

$$\frac{40}{10} = 4$$

From the histogram, the missing frequencies are

$$10 \leq d < 15 : 35, \quad 15 \leq d < 20 : 40, \quad 20 \leq d < 30 : 60$$

Together with the table:

$$40 + 35 + 40 + 60 + 30 = 205$$

For  $d > 25$ :

$$5(6) + 30 = 60$$

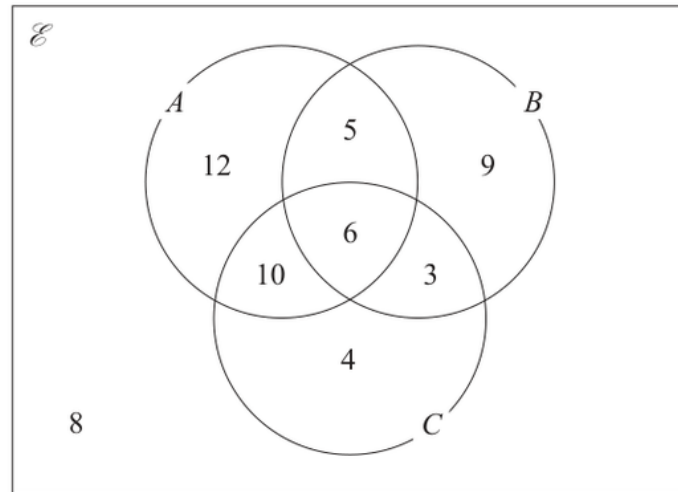
$$P(d > 25) = \frac{60}{205} = \frac{12}{41}$$

**FINAL ANSWER**completed frequencies 35, 40, 60, missing histogram density 1, and probability  $\frac{12}{41}$ .

**PAST PAPER QUESTION****Q#: 19****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

May 2021 P1H - May 2021 | P1H

19 The Venn diagram shows a universal set,  $\mathcal{E}$  and sets  $A$ ,  $B$  and  $C$ .



12, 5, 9, 10, 6, 3, 4 and 8 represent the **numbers** of elements.

Find

(i)  $n(A \cup B)$

.....  
(1)

(ii)  $n(A' \cap B')$

.....  
(1)

(iii)  $n([A \cap B] \cup C)$

.....  
(1)

**(Total for Question 19 is 3 marks)**

**PAST PAPER SOLUTION****Q#: 19****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Set notation and Venn diagrams. The tag is correct.**Method**

From the Venn diagram:

- $A$  only is 12
- $A \cap B$  only is 5
- $B$  only is 9
- $A \cap C$  only is 10
- all three is 6
- $B \cap C$  only is 3
- $C$  only is 4
- outside is 8

For  $A \cup B$ , add every region inside  $A$  or  $B$ :

$$12 + 5 + 9 + 10 + 6 + 3 = 45$$

For  $A' \cap B'$ , add the regions outside both  $A$  and  $B$ :

$$4 + 8 = 12$$

For  $(A \cap B) \cup C$ , add  $A \cap B$  and all of  $C$ :

$$5 + 6 + 10 + 3 + 4 = 28$$

**FINAL ANSWER**

(i) 45, (ii) 12, (iii) 28

**PAST PAPER QUESTION****Q#: 20****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May 2021 P1H - May 2021 | P1H

**20**  $P = \frac{t - w}{y}$

 $t = 9.7$  correct to 1 decimal place $w = 5.9$  correct to 1 decimal place $y = 3$  correct to 1 significant figure

Calculate the upper bound for the value of  $P$ .  
Show your working clearly.

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(Total for Question 20 is 3 marks)

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**PAST PAPER SOLUTION****Q#: 20****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Rounding, Estimation and Bounds. The tag is correct.**Method**

$$P = \frac{t - w}{y}$$

For the upper bound, make the numerator large and the denominator small.

$$t < 9.75, \quad w \geq 5.85, \quad y \geq 2.5$$

$$P_{\text{upper}} = \frac{9.75 - 5.85}{2.5} = 1.56$$

**FINAL ANSWER**

1.56

**PAST PAPER QUESTION****Q#: 20****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May 2021 P1H - May 2021 | P1H

**20**  $P = \frac{t - w}{y}$

 $t = 9.7$  correct to 1 decimal place $w = 5.9$  correct to 1 decimal place $y = 3$  correct to 1 significant figure

Calculate the upper bound for the value of  $P$ .  
Show your working clearly.

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(Total for Question 20 is 3 marks)

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**PAST PAPER SOLUTION****Q#: 20****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Rounding, Estimation & Bounds is correct.**Method**

$$P = \frac{t - w}{y}$$

For the upper bound of  $P$ , make the numerator as large as possible and the denominator as small as possible.

$$t = 9.75, \quad w = 5.85, \quad y = 2.5$$

$$P = \frac{9.75 - 5.85}{2.5}$$

$$P = \frac{3.90}{2.5} = 1.56$$

**FINAL ANSWER**

1.56

**PAST PAPER QUESTION****Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

May 2021 P1H - May 2021 | P1H

**21** Given that  $x = \frac{5}{9y+5}$  and that  $y = \frac{5}{5a-2}$

find an expression for  $x$  in terms of  $a$ .

Give your expression as a single fraction in its simplest form.

---

**(Total for Question 21 is 4 marks)**

Dr. Eslam Ahmed

**PAST PAPER SOLUTION****Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Corrected from Sine/Cosine Rule to Algebraic Fractions.**Method**

Given

$$x = \frac{5}{9y + 5}$$

and

$$y = \frac{5}{5a - 2}$$

Substitute  $y = \frac{5}{5a-2}$  into the expression for  $x$ :

$$x = \frac{5}{9\left(\frac{5}{5a-2}\right) + 5}$$

$$x = \frac{5}{\frac{45}{5a-2} + 5}$$

Write the denominator as a single fraction:

$$\begin{aligned} \frac{45}{5a-2} + 5 &= \frac{45 + 5(5a-2)}{5a-2} \\ &= \frac{45 + 25a - 10}{5a-2} \\ &= \frac{25a + 35}{5a-2} \end{aligned}$$

So

$$x = 5 \div \frac{25a + 35}{5a - 2}$$

$$x = 5 \times \frac{5a - 2}{25a + 35}$$

$$x = \frac{5(5a - 2)}{5(5a + 7)}$$

$$x = \frac{5a - 2}{5a + 7}$$

**FINAL ANSWER**

$$\frac{5a - 2}{5a + 7}$$

**PAST PAPER QUESTION****Q#: 21****Marks: 4**TOPIC: **Algebraic Fractions**

May 2021 P1H - May 2021 | P1H

**21** Given that  $x = \frac{5}{9y+5}$  and that  $y = \frac{5}{5a-2}$

find an expression for  $x$  in terms of  $a$ .

Give your expression as a single fraction in its simplest form.

---

(Total for Question 21 is 4 marks)

Dr. Eslam Ahmed

**PAST PAPER SOLUTION****Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Corrected from Sine/Cosine Rule to Algebraic Fractions.**Method**

Given

$$x = \frac{5}{9y + 5}$$

and

$$y = \frac{5}{5a - 2}$$

Substitute  $y = \frac{5}{5a-2}$  into the expression for  $x$ :

$$x = \frac{5}{9\left(\frac{5}{5a-2}\right) + 5}$$

$$x = \frac{5}{\frac{45}{5a-2} + 5}$$

Write the denominator as a single fraction:

$$\begin{aligned} \frac{45}{5a-2} + 5 &= \frac{45 + 5(5a-2)}{5a-2} \\ &= \frac{45 + 25a - 10}{5a-2} \\ &= \frac{25a + 35}{5a-2} \end{aligned}$$

So

$$x = 5 \div \frac{25a + 35}{5a - 2}$$

$$x = 5 \times \frac{5a - 2}{25a + 35}$$

$$x = \frac{5(5a - 2)}{5(5a + 7)}$$

$$x = \frac{5a - 2}{5a + 7}$$

**FINAL ANSWER**

$$\frac{5a - 2}{5a + 7}$$

**PAST PAPER QUESTION****Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

May 2021 P1H - May 2021 | P1H

- 22** The diagram shows a triangular prism  $ABCDEF$  with a horizontal base  $ABEF$ .

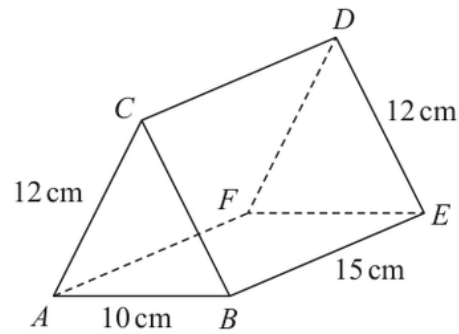


Diagram **NOT**  
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between  $AD$  and the base  $ABEF$ .  
Give your answer correct to 3 significant figures.

.....°

(Total for Question 22 is 4 marks)



**PAST PAPER SOLUTION****Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The triangular end is isosceles, with

$$AC = BC = 12, \quad AB = 10$$

The perpendicular height of the triangle is found by halving  $AB$ :

$$AM = 5$$

$$CM^2 = 12^2 - 5^2 = 144 - 25 = 119$$

$$CM = \sqrt{119}$$

The projection of  $AD$  on the horizontal base goes from  $A$  to the midpoint of  $FE$ . Its length is

$$\sqrt{15^2 + 5^2} = \sqrt{250}$$

So, for the angle  $\theta$  between  $AD$  and the base,

$$\tan \theta = \frac{\sqrt{119}}{\sqrt{250}}$$

$$\theta = \tan^{-1} \left( \frac{\sqrt{119}}{\sqrt{250}} \right) = 34.602 \dots$$

**FINAL ANSWER** $34.6^\circ$

**PAST PAPER QUESTION****Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

May 2021 P1H - May 2021 | P1H

- 22** The diagram shows a triangular prism  $ABCDEF$  with a horizontal base  $ABEF$ .

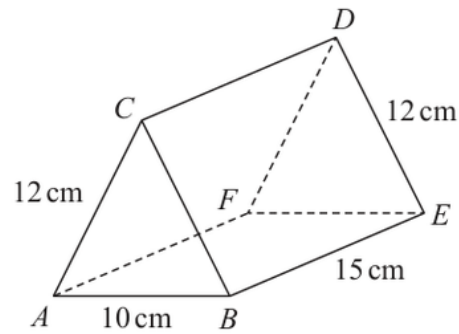


Diagram **NOT**  
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$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between  $AD$  and the base  $ABEF$ .  
Give your answer correct to 3 significant figures.

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(Total for Question 22 is 4 marks)

**PAST PAPER SOLUTION****Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

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$$\tan \theta = \frac{\sqrt{119}}{\sqrt{250}}$$

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**FINAL ANSWER**

34.6°

**PAST PAPER QUESTION****Q#: 23****Marks: 5**TOPIC: **Sequences**

May 2021 P1H - May 2021 | P1H

- 23** The sum of the first  $N$  terms of an arithmetic series,  $S$ , is 292  
The 2nd term of  $S$  is 8.5  
The 5th term of  $S$  is 13

Find the value of  $N$ .

Show clear algebraic working.

 $N = \dots\dots\dots$ 

---

**(Total for Question 23 is 5 marks)**

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Dr. Eslam Ahmed

**PAST PAPER SOLUTION****Q#: 23****Marks: 5**TOPIC: **Sequences**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The 2nd term is

$$a + d = 8.5$$

The 5th term is

$$a + 4d = 13$$

Subtract:

$$3d = 4.5$$

$$d = 1.5$$

$$a = 8.5 - 1.5 = 7$$

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$292 = \frac{N}{2}(14 + 1.5(N - 1))$$

$$292 = \frac{N}{2}(1.5N + 12.5)$$

$$3N^2 + 25N - 1168 = 0$$

$$N = \frac{-25 + 121}{6} = 16$$

**FINAL ANSWER**

$$N = 16$$

**PAST PAPER QUESTION****Q#: 23****Marks: 5**TOPIC: **Sequences**

May 2021 P1H - May 2021 | P1H

- 23** The sum of the first  $N$  terms of an arithmetic series,  $S$ , is 292  
The 2nd term of  $S$  is 8.5  
The 5th term of  $S$  is 13

Find the value of  $N$ .

Show clear algebraic working.

 $N = \dots\dots\dots$ 

---

**(Total for Question 23 is 5 marks)**

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Dr. Eslam Ahmed

**PAST PAPER SOLUTION****Q#: 23****Marks: 5**TOPIC: **Sequences**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The 2nd term is

$$a + d = 8.5$$

The 5th term is

$$a + 4d = 13$$

Subtract:

$$3d = 4.5$$

$$d = 1.5$$

$$a = 8.5 - 1.5 = 7$$

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$292 = \frac{N}{2}(14 + 1.5(N - 1))$$

$$292 = \frac{N}{2}(1.5N + 12.5)$$

$$3N^2 + 25N - 1168 = 0$$

$$N = \frac{-25 + 121}{6} = 16$$

**FINAL ANSWER**

$$N = 16$$

**PAST PAPER QUESTION****Q#: 24****Marks: 6****TOPIC: Functions**

May 2021 P1H - May 2021 | P1H

**24** The functions  $f$  and  $g$  are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find  $fg(2)$ .....  
(2)(b) Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$ 

$$f^{-1}(x) = \dots$$

(4)

**(Total for Question 24 is 6 marks)**



**PAST PAPER SOLUTION****Q#: 24****Marks: 6****TOPIC: Functions**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

(a)

$$g(2) = 7(2) - 6 = 8$$

$$fg(2) = f(8)$$

$$f(8) = 5(8)^2 - 10(8) + 7$$

$$f(8) = 247$$

(b)

$$f(x) = 5x^2 - 10x + 7$$

$$f(x) = 5(x - 1)^2 + 2$$

Let

$$y = 5(x - 1)^2 + 2$$

$$(x - 1)^2 = \frac{y - 2}{5}$$

Since the domain is  $x \geq 1$ , take the positive square root:

$$x = 1 + \sqrt{\frac{y - 2}{5}}$$

**FINAL ANSWER**

$$(a) 247, (b) f^{-1}(x) = 1 + \sqrt{\frac{x - 2}{5}}$$

**PAST PAPER QUESTION****Q#: 24****Marks: 6****TOPIC: Functions**

May 2021 P1H - May 2021 | P1H

**24** The functions  $f$  and  $g$  are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find  $fg(2)$ .....  
(2)(b) Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$ 

$$f^{-1}(x) = \dots$$

(4)

**(Total for Question 24 is 6 marks)**

**PAST PAPER SOLUTION****Q#: 24****Marks: 6****TOPIC: Functions**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

(a)

$$g(2) = 7(2) - 6 = 8$$

$$fg(2) = f(8)$$

$$f(8) = 5(8)^2 - 10(8) + 7$$

$$f(8) = 247$$

(b)

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$$f(x) = 5(x - 1)^2 + 2$$

Let

$$y = 5(x - 1)^2 + 2$$

$$(x - 1)^2 = \frac{y - 2}{5}$$

Since the domain is  $x \geq 1$ , take the positive square root:

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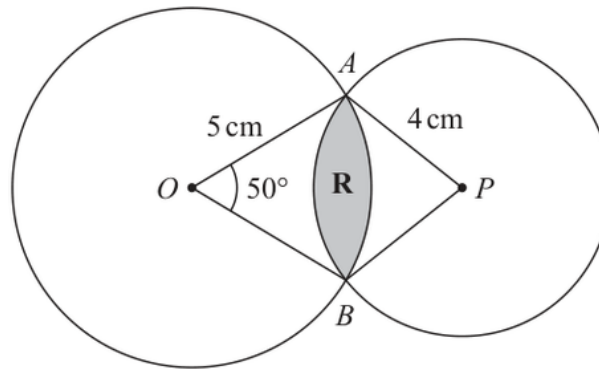
**FINAL ANSWER**

$$(a) 247, (b) f^{-1}(x) = 1 + \sqrt{\frac{x - 2}{5}}$$

**PAST PAPER QUESTION****Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

- 25** The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

Diagram **NOT**  
accurately drawn

One of the circles has centre  $O$  and radius  $5\text{ cm}$ .  
 The other circle has centre  $P$  and radius  $4\text{ cm}$ .  
 Angle  $AOB = 50^\circ$

Calculate the area of region **R**.  
 Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$ **(Total for Question 25 is 6 marks)**

**PAST PAPER SOLUTION****Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Moved to Circles, Arcs & Sectors.**Method**For the circle with centre  $O$ :

$$\angle AOB = 50^\circ, \quad OA = OB = 5$$

The half chord length is

$$\frac{AB}{2} = 5 \sin 25^\circ$$

For the circle with centre  $P$ , let

$$\angle APB = \theta$$

The radius is 4, so

$$4 \sin \left( \frac{\theta}{2} \right) = 5 \sin 25^\circ$$

$$\sin \left( \frac{\theta}{2} \right) = \frac{5 \sin 25^\circ}{4}$$

$$\theta = 63.7777 \dots^\circ$$

The common region is made from two minor segments.

For the circle with centre  $O$ :

$$\text{segment area} = \frac{50}{360} \pi (5)^2 - \frac{1}{2} (5)^2 \sin 50^\circ = 1.3328 \dots$$

For the circle with centre  $P$ :

$$\begin{aligned} \text{segment area} &= \frac{\theta}{360} \pi (4)^2 - \frac{1}{2} (4)^2 \sin \theta \\ &= 1.7284 \dots \end{aligned}$$

Total area of region  $R$ :

$$1.3328 \dots + 1.7284 \dots = 3.0611 \dots$$

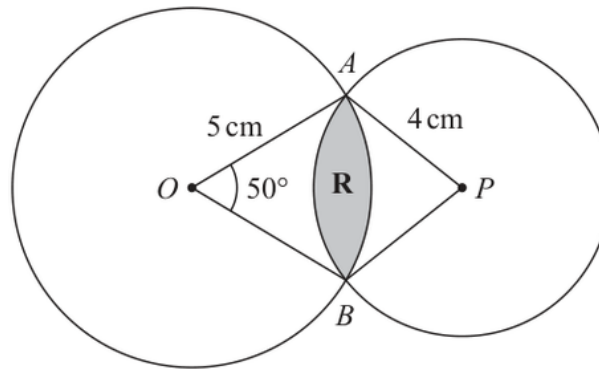
**FINAL ANSWER**

$$\boxed{3.06 \text{ cm}^2}$$

**PAST PAPER QUESTION****Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

- 25** The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

Diagram **NOT**  
accurately drawn

One of the circles has centre  $O$  and radius 5 cm.  
 The other circle has centre  $P$  and radius 4 cm.  
 Angle  $AOB = 50^\circ$

Calculate the area of region **R**.  
 Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$ 

(Total for Question 25 is 6 marks)

**PAST PAPER SOLUTION****Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Moved to Circles, Arcs & Sectors.**Method**For the circle with centre  $O$ :

$$\angle AOB = 50^\circ, \quad OA = OB = 5$$

The half chord length is

$$\frac{AB}{2} = 5 \sin 25^\circ$$

For the circle with centre  $P$ , let

$$\angle APB = \theta$$

The radius is 4, so

$$4 \sin \left( \frac{\theta}{2} \right) = 5 \sin 25^\circ$$

$$\sin \left( \frac{\theta}{2} \right) = \frac{5 \sin 25^\circ}{4}$$

$$\theta = 63.7777 \dots^\circ$$

The common region is made from two minor segments.

For the circle with centre  $O$ :

$$\text{segment area} = \frac{50}{360} \pi (5)^2 - \frac{1}{2} (5)^2 \sin 50^\circ = 1.3328 \dots$$

For the circle with centre  $P$ :

$$\begin{aligned} \text{segment area} &= \frac{\theta}{360} \pi (4)^2 - \frac{1}{2} (4)^2 \sin \theta \\ &= 1.7284 \dots \end{aligned}$$

Total area of region  $R$ :

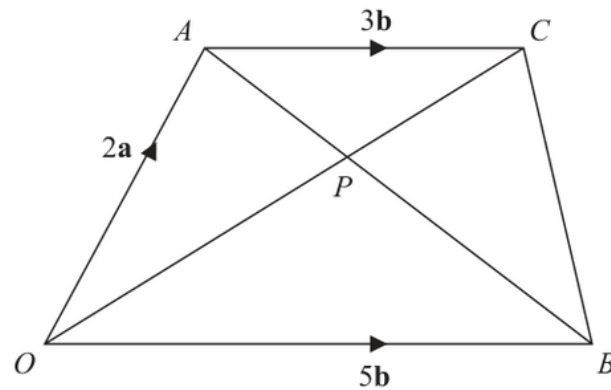
$$1.3328 \dots + 1.7284 \dots = 3.0611 \dots$$

**FINAL ANSWER**

$$\boxed{3.06 \text{ cm}^2}$$

**PAST PAPER QUESTION****Q#: 26****Marks: 5**TOPIC: **Vectors**

May 2021 P1H - May 2021 | P1H

**26**  $OACB$  is a trapezium.Diagram **NOT**  
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 5\mathbf{b} \quad \vec{AC} = 3\mathbf{b}$$

The diagonals,  $OC$  and  $AB$ , of the trapezium intersect at the point  $P$ .Find and simplify an expression, in terms of  $\mathbf{a}$  and  $\mathbf{b}$ , for  $\vec{OP}$ 

Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

**(Total for Question 26 is 5 marks)**



**PAST PAPER SOLUTION****Q#: 26****Marks: 5**TOPIC: **Vectors**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Moved to Vectors.**Method**

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 5\mathbf{b}, \quad \overrightarrow{AC} = 3\mathbf{b}$$

So

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = 2\mathbf{a} + 3\mathbf{b}$$

Since  $P$  lies on  $OC$ , let

$$\overrightarrow{OP} = t(2\mathbf{a} + 3\mathbf{b})$$

Since  $P$  also lies on  $AB$ , let

$$\overrightarrow{OP} = \overrightarrow{OA} + s\overrightarrow{AB}$$

Now

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 5\mathbf{b} - 2\mathbf{a}$$

So

$$\overrightarrow{OP} = 2\mathbf{a} + s(5\mathbf{b} - 2\mathbf{a})$$

$$\overrightarrow{OP} = (2 - 2s)\mathbf{a} + 5s\mathbf{b}$$

Compare this with

$$t(2\mathbf{a} + 3\mathbf{b}) = 2t\mathbf{a} + 3t\mathbf{b}$$

Equating coefficients:

$$2t = 2 - 2s$$

$$3t = 5s$$

From  $t = 1 - s$ ,

$$3(1 - s) = 5s$$

$$3 = 8s$$

$$s = \frac{3}{8}$$

So

$$t = 1 - \frac{3}{8} = \frac{5}{8}$$

Therefore

$$\overrightarrow{OP} = \frac{5}{8}(2\mathbf{a} + 3\mathbf{b})$$

$$\overrightarrow{OP} = \frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

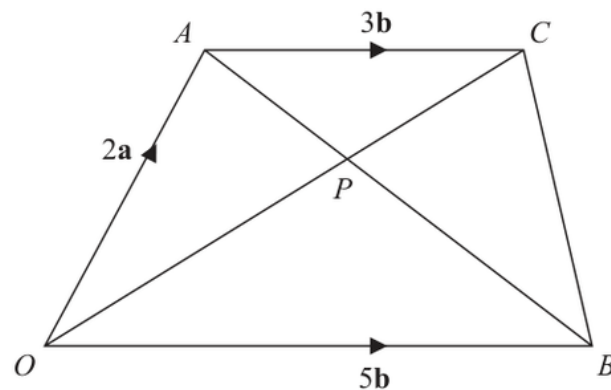
**FINAL ANSWER**

$$\frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

Dr. Eslam Ahmed

**PAST PAPER QUESTION****Q#: 26****Marks: 5**TOPIC: **Vectors**

May 2021 P1H - May 2021 | P1H

**26**  $OACB$  is a trapezium.Diagram **NOT**  
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 5\mathbf{b} \quad \vec{AC} = 3\mathbf{b}$$

The diagonals,  $OC$  and  $AB$ , of the trapezium intersect at the point  $P$ .Find and simplify an expression, in terms of  $\mathbf{a}$  and  $\mathbf{b}$ , for  $\vec{OP}$ 

Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

**(Total for Question 26 is 5 marks)**

**PAST PAPER SOLUTION****Q#: 26****Marks: 5**TOPIC: **Vectors**

May 2021 P1H - May 2021 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Moved to Vectors.**Method**

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 5\mathbf{b}, \quad \overrightarrow{AC} = 3\mathbf{b}$$

So

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = 2\mathbf{a} + 3\mathbf{b}$$

Since  $P$  lies on  $OC$ , let

$$\overrightarrow{OP} = t(2\mathbf{a} + 3\mathbf{b})$$

Since  $P$  also lies on  $AB$ , let

$$\overrightarrow{OP} = \overrightarrow{OA} + s\overrightarrow{AB}$$

Now

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 5\mathbf{b} - 2\mathbf{a}$$

So

$$\overrightarrow{OP} = 2\mathbf{a} + s(5\mathbf{b} - 2\mathbf{a})$$

$$\overrightarrow{OP} = (2 - 2s)\mathbf{a} + 5s\mathbf{b}$$

Compare this with

$$t(2\mathbf{a} + 3\mathbf{b}) = 2t\mathbf{a} + 3t\mathbf{b}$$

Equating coefficients:

$$2t = 2 - 2s$$

$$3t = 5s$$

From  $t = 1 - s$ ,

$$3(1 - s) = 5s$$

$$3 = 8s$$

$$s = \frac{3}{8}$$

So

$$t = 1 - \frac{3}{8} = \frac{5}{8}$$

Therefore

$$\overrightarrow{OP} = \frac{5}{8}(2\mathbf{a} + 3\mathbf{b})$$

$$\overrightarrow{OP} = \frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

**FINAL ANSWER**

$$\frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

Dr. Eslam Ahmed